

Toolbar Usage Documentation

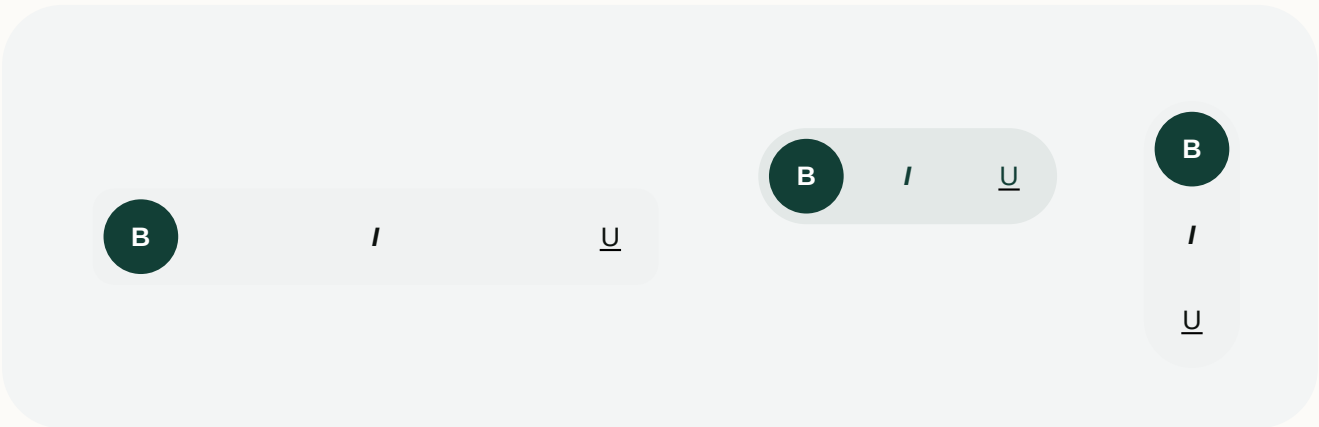
Implementation, design behavior, accessibility, and Figma handoff for Technical Director and Designer review.

REVIEW STATUS

Implementation-aligned reference. The Angular component, documentation playground, and Figma component set were validated together on 21 September 2026.

6	4	3	0
Figma variants	Angular inputs	Content slots	FAB/Create APIs

DOCUMENT OWNER	AI Labs Design System
AUDIENCE	Technical Directors, Product Designers, Front-end Engineers
SOURCE	Angular library + Figma component set



What the Toolbar is for

A focused action container for commands that are relevant to the current page or selection. It supports docked and floating placement without taking ownership of the action logic.

Approved application

- Place frequently used, closely related actions near the work they affect.
- Use Docked when the toolbar belongs to the page edge or a persistent work area.
- Use Floating Horizontal for compact contextual actions with available width.
- Use Floating Vertical when horizontal space is constrained or the work surface benefits from a side-aligned control strip.
- Use Vibrant only when the action group needs stronger visual emphasis than the surrounding surface.

Do not use

- Do not use the Toolbar as primary navigation or show it simultaneously with a navigation bar in the same region.
- Do not add a Create button or FAB inside this component. That option was intentionally removed from both code and Figma.
- Do not place unrelated commands together merely to fill available slots.
- Do not depend on icons without accessible labels, tooltips, or contextual clarity.

DECISION SUMMARY

The component intentionally remains a flexible container. Actions are projected by the application, while the Toolbar owns layout, tone, orientation, responsive spacing, and expanded/collapsed visibility.

Implementation status

Area	Status	Review note
Angular	Complete	Standalone component, OnPush change detection, typed variants and tones.
Figma	Complete	Six token-bound variants with one boolean property: Expanded.
Material icons	Complete	format_bold, format_italic, format_underlined at 24px.
Create/FAB	Removed	No input, projection slot, control, layer, or usage guidance remains.
Validation	Passed	Production build plus visual checks for docked, vertical, vibrant, and collapsed states.

One container, three action regions

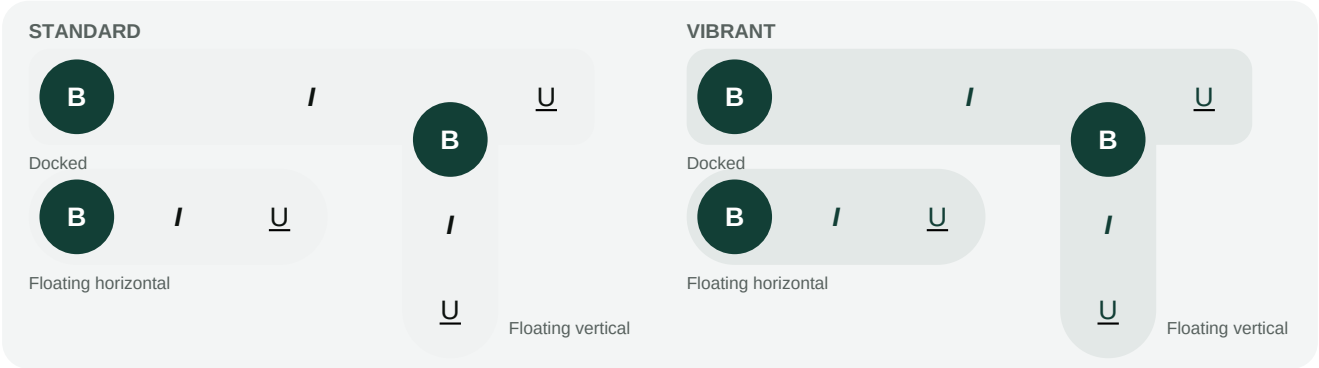
The Toolbar is composed of a 64px container and three projected regions. The application supplies the controls; the component keeps their placement and state behavior consistent.

Region	Angular projection	Purpose	Collapsed behavior
Leading	toolbarLeading	First contextual action or selected toggle.	Hidden on floating variants.
Primary	Default ng-content	The action retained when a floating toolbar collapses.	Always retained.
Trailing	toolbarTrailing	Last contextual action or secondary command.	Hidden on floating variants.

Variant matrix

Variant	Standard	Vibrant	Expanded
Docked	Available	Available	All three regions remain visible; Expanded does not hide docked content.
Floating Horizontal	Available	Available	Leading and trailing are visible when true; only primary remains when false.
Floating Vertical	Available	Available	Uses the same visibility rule with vertical orientation.

Validated Figma variants



Validated component-set structure: Docked, Floating Horizontal, and Floating Vertical in Standard and Vibrant tones.

DESIGNER NOTE

The selected Bold action uses the brand container and inverse icon roles. Unselected actions use icon/primary in Standard and icon/brand in Vibrant.

03 / VISUAL SPECIFICATION

Measurements and semantic tokens

Measurements follow the current implementation and Material toolbar guidance: a 64px container, 48px action targets, 24px icons, centered alignment, and token-driven color.

Property	Value	Implementation
Container height	64px minimum	.ds-toolbar__container min-height
Action target	48 x 48px	All toolbar Button actions
Icon size	24 x 24px	.material-icon
Item gap	8px	--spacing-8
Docked horizontal padding	16px	--spacing-16; reduces to 8px below 520px
Floating padding	8px	Compact outside padding
Docked radius	16px top corners	--radius-xl
Floating radius	100px pill	--radius-pill
Floating elevation	Small shadow	--shadow-sm

Tone mapping

Role	Standard	Vibrant	Selected action
Container	background/tertiary	background/brand-subtle	background/brand
Unselected icons	icon/primary	icon/brand	Not applicable
Selected icon	icon/inverse	icon/inverse	icon/inverse
Dark mode	Semantic alias resolves	Semantic alias resolves	Semantic alias resolves

Material icon mapping

Position	Material icon	Example meaning	Treatment
Leading	format_bold	Bold formatting	Selected, circular brand fill
Primary	format_italic	Italic formatting	Ghost icon button
Trailing	format_underlined	Underline formatting	Ghost icon button

ICON RULE

Use official Material Icons or Material Symbols artwork. Keep the 24px viewBox, use `currentColor` through the `material-icon` class, and provide a unique accessible name for every icon-only action.

Component contract

ToolbarComponent is standalone and presentation-only. It accepts layout and accessibility inputs, and relies on projected Button controls for actions and event handlers.

Input	Type	Default	Behavior
variant	ToolbarVariant	docked	docked floating-horizontal floating-vertical
tone	ToolbarTone	standard	standard vibrant
expanded	boolean	true	Controls auxiliary regions on floating variants.
ariaLabel	string	Page actions	Accessible name applied to role=toolbar.

Public types

```
export type ToolbarVariant =
  | 'docked'
  | 'floating-horizontal'
  | 'floating-vertical'

export type ToolbarTone = 'standard' | 'vibrant'
```

Recommended template

```
<ds-toolbar
  variant="floating-horizontal"
  tone="vibrant"
  [expanded]="toolbarExpanded"
  ariaLabel="Text formatting"
>
  <button dsButton toolbarLeading [iconOnly]="true" aria-label="Bold">
    <svg class="material-icon" aria-hidden="true">
      <use href="#icon-material-format-bold"></use>
    </svg>
  </button>
  <button dsButton variant="ghost" [iconOnly]="true" aria-label="Italic">...</button>
  <button dsButton toolbarTrailing variant="ghost" [iconOnly]="true"
    aria-label="Underline">...</button>
</ds-toolbar>
```

Import

```
import { ButtonComponent, ToolbarComponent } from './design-system'

@Component({
  standalone: true,
  imports: [ButtonComponent, ToolbarComponent],
})
```

State ownership and responsive use

The consuming feature owns action state, click handlers, pressed state, availability, and toolbar expansion. The design-system component does not infer business logic.

State ownership

- Bind expanded to application state when the toolbar must collapse based on workspace mode, viewport, or selection context.
- Use aria-pressed on toggle actions such as Bold, Italic, and Underline. Do not use visual selection alone.
- Disable actions only when they are unavailable; preserve the toolbar and explain restrictions through adjacent context or a tooltip.
- Keep primary action ordering stable as state changes. Reordering controls creates avoidable motor and cognitive cost.

Collapsed behavior

Variant	expanded=true	expanded=false
Docked	Leading + Primary + Trailing	No structural change
Floating Horizontal	Leading + Primary + Trailing	Primary only; container hugs content
Floating Vertical	Leading + Primary + Trailing	Primary only; container hugs content

Responsive behavior

- Below 520px, Docked horizontal padding changes from 16px to 8px and group gaps reduce to 4px.
- Floating variants remain content-sized and should be positioned by the consuming layout, not by internal absolute positioning.
- Use Horizontal where three targets fit without covering critical content. Use Vertical when it improves reach or preserves the work surface.

Framework responsibility boundary

Toolbar owns	Application owns
Container layout, tone, orientation, spacing, collapsed projection	Action labels, click handlers, pressed state, disabled state, business permissions
Semantic toolbar role and orientation	Context-specific ariaLabel and focus strategy
Visual integration with design tokens	Placement, visibility trigger, persistence, analytics

REMOVED API

showFab, toolbarFab, Show FAB, and the Create action are not part of the supported contract. Do not reintroduce them in downstream templates or Figma instances.

Interaction and assistive technology requirements

The base component provides a semantic toolbar and orientation. Product teams must complete accessible naming, toggle state, focus behavior, and keyboard behavior for the projected controls.

Requirement	Implementation guidance	Owner
Accessible name	Set ariaLabel to the action group purpose, for example Text formatting.	Application
Icon-only labels	Every icon button requires an aria-label; SVGs should be aria-hidden.	Application
Orientation	horizontal for Docked/Horizontal; vertical for Floating Vertical.	Toolbar
Touch target	Maintain the 48 x 48px action target.	Toolbar
Keyboard	Tab reaches each enabled button. Enter and Space activate controls.	Button + browser
Toggle state	Expose aria-pressed for formatting toggles.	Application
Focus visibility	Retain the Button focus ring; never suppress it without a replacement.	Button
Color	Use semantic tokens and verify contrast in both Light and Dark themes.	Tokens + QA
Collapsed state	Do not move focus into an action immediately before hiding that action.	Application

Keyboard model

- Minimum supported model: normal Tab sequence across projected buttons.
- If the product adopts roving tabindex and arrow-key navigation, implement it consistently across all toolbars and document the pattern separately.
- Escape may close a contextual floating toolbar only when the toolbar is presented as a dismissible transient layer.
- Restoring focus after dismissal is the responsibility of the feature that opened the toolbar.

Review tests

Test	Expected result
Screen reader	Toolbar purpose, orientation, action names, and pressed states are announced.
Keyboard only	All visible actions are reachable and focus is never trapped or lost.
200% zoom	Floating and docked layouts remain usable without covering essential content.
Dark mode	Container, selected action, icons, hover, and focus remain distinguishable.
Reduced width	Docked spacing compresses; floating placement remains application-controlled.

Variant and token contract

The Figma component mirrors the code API. Designers should use component properties and semantic variables rather than detaching instances or applying hardcoded color.

Figma field	Values	Code mapping
Variant	Docked Floating Horizontal Floating Vertical	variant input
Tone	Standard Vibrant	tone input
Expanded	true false	expanded input
Layer: leading	Material format_bold example	toolbarLeading projection
Layer: primary	Material format_italic example	default projection
Layer: trailing	Material format_underlined example	toolbarTrailing projection

Component set identity

Item	Reference
Figma file	AILab Design System / FZDQAzD6Q44OMCeYDKdfp8
Toolbar page	Node 63:2
Toolbar component set	Node 68:2
Variant count	6
Boolean properties	Expanded only

Token bindings

Element	Standard	Vibrant
Container	background/tertiary	background/brand-subtle
Selected action	background/brand	background/brand
Selected icon	icon/inverse	icon/inverse
Other icons	icon/primary	icon/brand

Designer handoff rules

- Keep instances attached to the component set. Override content through the supported action regions.
- Do not add a fourth Create/FAB action to the component. Use a separate application-level action if product requirements demand one.
- Preserve 48px targets and 24px icon geometry when changing action artwork.
- Use Standard by default; use Vibrant deliberately for stronger emphasis.
- Annotate product-specific behavior, visibility rules, and toggle states in the consuming screen, not in the base component.

Architecture, verification, and acceptance

The component is intentionally small: a typed Angular container with OnPush change detection, projected controls, semantic CSS, and no business events of its own.

Source locations

Purpose	Repository path
Component API	src/app/design-system/toolbar.component.ts
Component export	src/app/design-system/index.ts
Component styling	src/design-system/components.css
Semantic tokens	src/design-system/tokens.css
Catalog and playground	src/app/app.component.html and app.component.ts

Architecture notes

- ChangeDetectionStrategy.OnPush limits unnecessary checks for this presentation component.
- Host classes encode variant, tone, and expanded state without runtime style generation.
- Content projection keeps action semantics and event handling in the feature that owns them.
- Semantic CSS variables provide Light and Dark theme adaptation without component branching.
- No external icon font or network request is required; official Material SVG paths are embedded in the site sprite.

Verification completed

Check	Result	Evidence
Angular production build	Pass	ng build completed successfully on 21 September 2026.
Site visual QA	Pass	Docked, Floating Vertical Vibrant, and collapsed states inspected.
Figma metadata	Pass	Six variants; Variant, Tone, and Expanded properties only.
Figma visual QA	Pass	All 18 icon vectors visible; no Create/FAB content or clipping.
Variable audit	Pass	Container, selected action, and icon fills use semantic bindings.

Required automated coverage

- Render all variant and tone combinations and assert the toolbar role and orientation.
- Verify Docked ignores collapsed visibility while both floating variants hide leading and trailing regions.
- Verify custom ariaLabel is reflected on the toolbar container.
- Add keyboard and accessibility checks at the application pattern level because action logic is projected.
- Include screenshot regression coverage for Light and Dark themes if the project adds a visual-test runner.

Review checklist and source references

Use this page as the sign-off reference. Approval should confirm the component contract, visual behavior, accessibility expectations, and ongoing code-to-Figma parity.

Technical Director review

- Component scope is appropriate: presentation and layout only, with no hidden business logic.
- Public API is stable and minimal: variant, tone, expanded, ariaLabel, and three projection regions.
- The removed Create/FAB API will not be reintroduced without a new design-system decision.
- Accessibility responsibility is clearly split between the component and consuming feature.
- Build, test, release, and future visual-regression expectations are accepted.

Designer review

- Docked and floating forms are appropriate for intended product surfaces.
- Standard and Vibrant tones are used consistently with hierarchy.
- Material icons, selected state, target sizes, and spacing match the approved pattern.
- Figma variants and component properties match the Angular contract.
- Screen-level annotations cover product behavior not owned by the base component.

Change record

Version	Date	Change
1.0	21 Sep 2026	Initial review document after Create/FAB removal and Material icon alignment.

Primary references

- Material 3 Toolbars - Overview
- Material 3 Toolbars - Specifications
- Google Material Design Icons repository
- AILab Design System - Toolbar component set

SOURCE OF TRUTH

Production behavior is defined by the Angular component and semantic CSS. Figma is the design contract and must remain structurally aligned. When either changes, update both and revise this document's change record.

Approval	Name / role	Date	Decision
Technical			Approved / changes requested
Design			Approved / changes requested